

Deep Learning for Document Understanding

1. Introduction

Document AI converts unstructured PDFs into structured, editable formats. This sample exercises text, headings, tables, figures and mathematics.

2. The Loss Function

We minimise the cross-entropy objective shown below, where the sum runs over all tokens.

$$L(\theta) = - \sum_i y_i \log(p_i) + \lambda \sum_j \theta_j^2$$

$$\int_0^\infty e^{-x^2} dx = \sqrt{\pi}/2$$

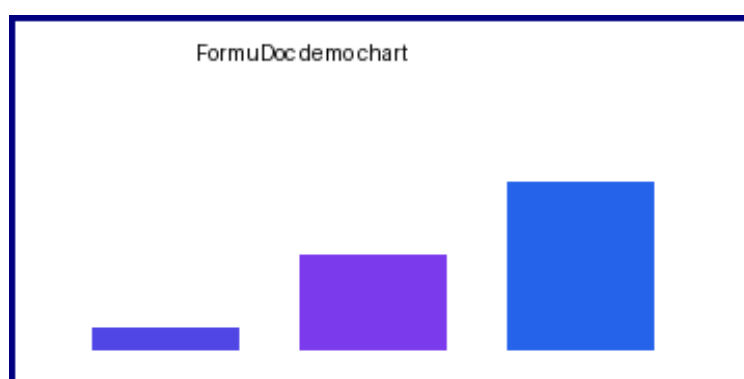


Figure 1: Training throughput by model size.

3. Results

Table 1 reports accuracy across configurations.

Table 1: Model comparison.

Model	Params	Accuracy
Baseline	12M	88.1%
FormuNet	45M	93.7%
FormuNet-XL	120M	95.2%